Experimental Determination of Einstein's Theory of Light and Planck's Constant with the Photoelectric Effect

Ethan Suresh, Vicki Mu, J-Lab Team

Two conflicting theories of light

Classical:

- Light is a continuous wave

$$\mathbf{E}(\mathbf{r}, t) = E_0\,\hat{\mathbf{x}}\cos(kz - \omega t + \phi),$$

$$I \propto |E_0|^2$$

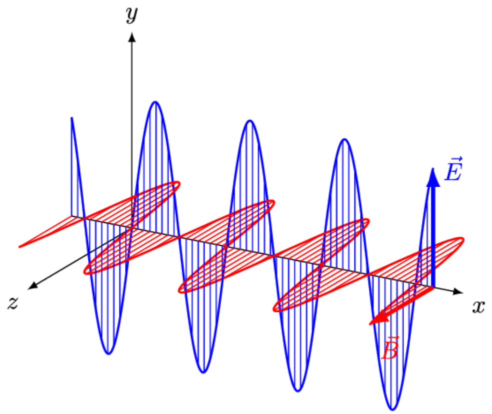


Image Source: https://commons.wikimedia.org/wiki/File:EM-Wave.gif

Einstein (Quantum)

- Light comes in quantized energy "photons"

$$E = h\nu$$

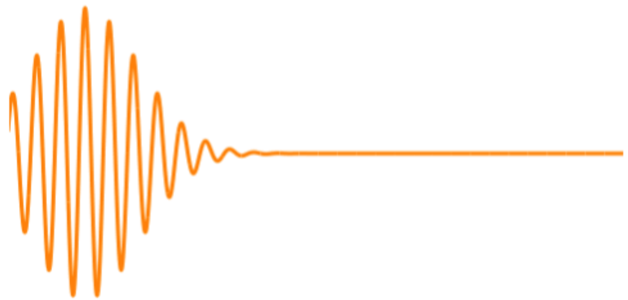
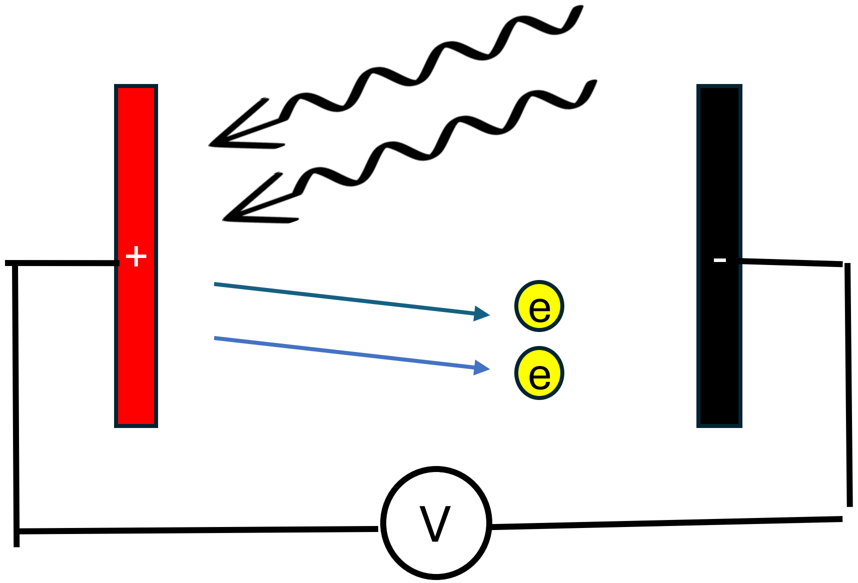


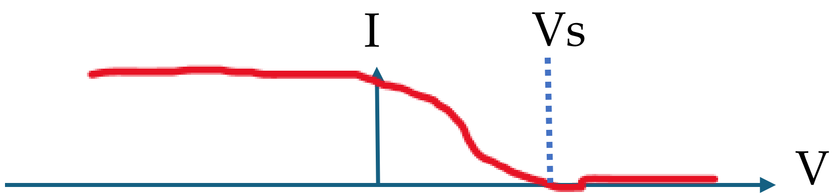
Image Source: https://commons.wikimedia.org/wiki/File:EM-Wave.gif

CITE IMAGE SOURCES

CITE IMAGE SOURCES

The Photoelectric Effect seems to support Einstein





- **Work Function ϕ:** "minimum" energy to eject an electron from a material

- Certain wavelengths of light ejects electrons from the material, overcoming the work function

$$K = h\nu - \phi$$

$$eV_s = h\nu - \phi_A$$

Source: Experiments in Modern Physics, A. Melissinos

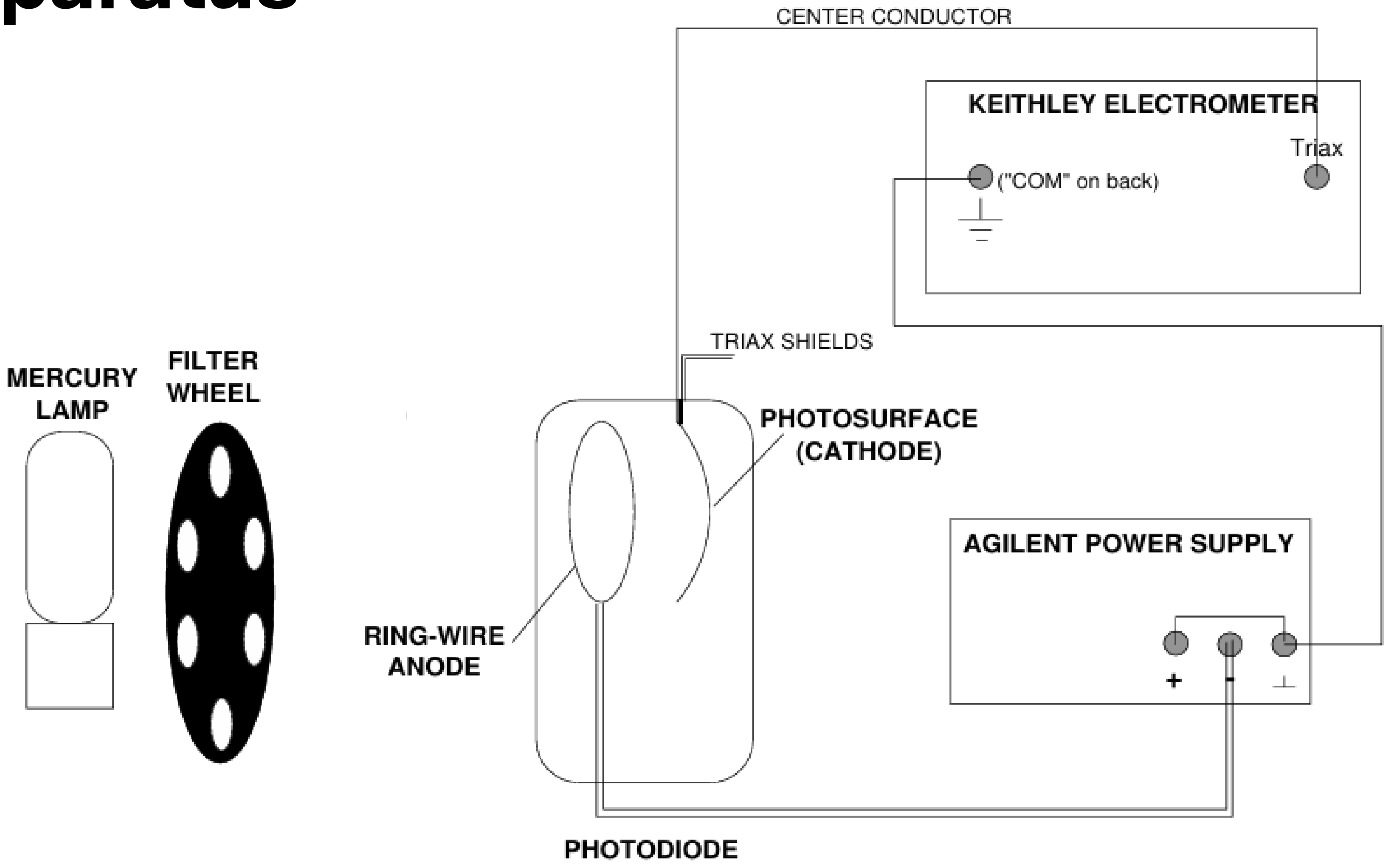
Experimental Objectives: Testing the Photoelectric Effect (PE)

We test Einstein's theory of light quanta against the classical theory of light using the Photoelectric effect. Additionally, we will measure Planck's constant and the work function of the anode.

- How does the photocurrent (counting number of electrons ejected from the cathode) depend on the applied voltage and wavelength?
- Is there a linear relationship between photon energy and frequency?
- What sources of systematic and statistical error confound our results?

Experimental Apparatus

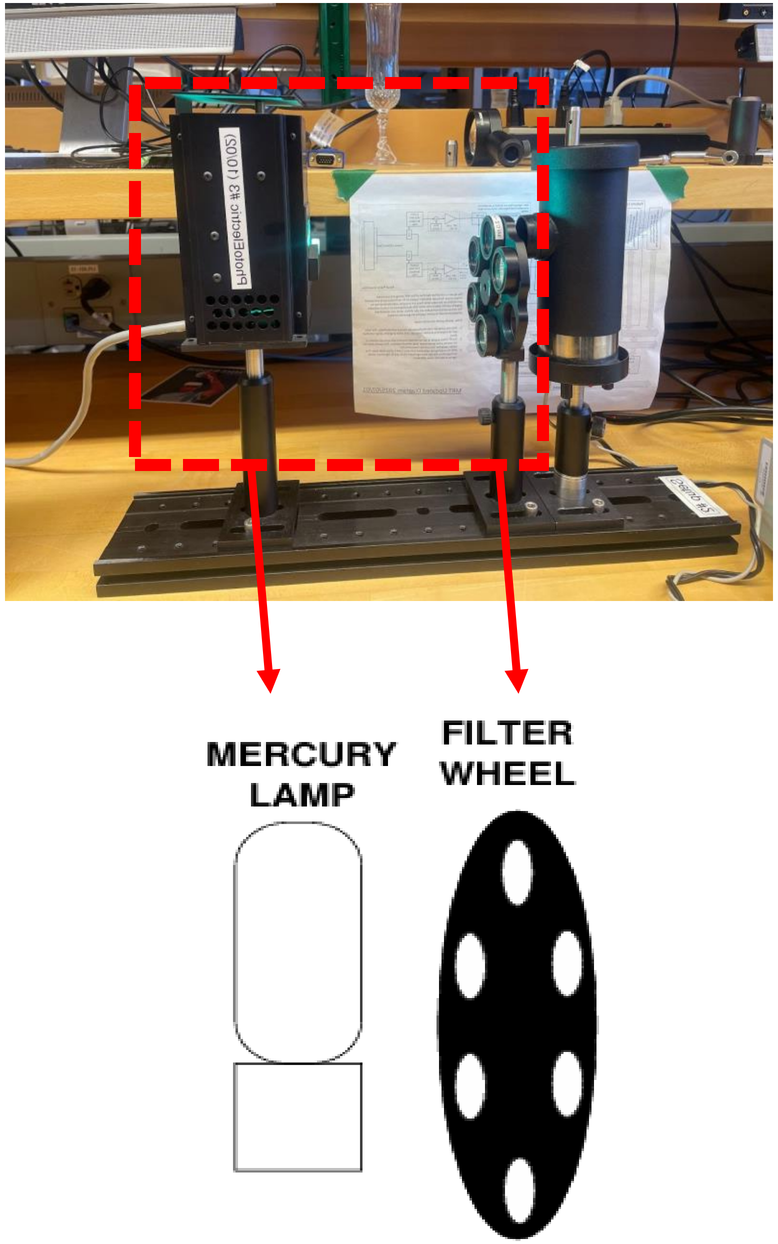
PE Apparatus



Source: *The Photoelectric Effect,* MIT Department of Physics

PE Apparatus: Oriel 65130 Mercury Lamp and Filter

- Provides high-intensity, sharp spectral lines
- Lamp is enclosed in a burner and housing case; requires a warmup period of ~ 10-15 min
- Light emitted from housing aperture

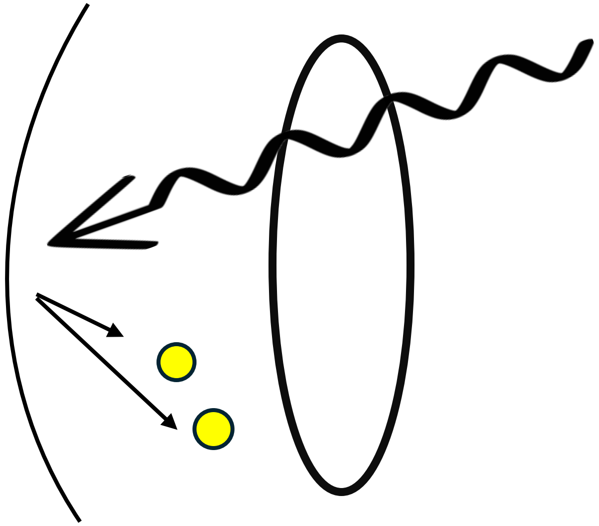


- **Frequencies Selected: 404.7, 435.8, 577.0**
  - **Uncertainty:** 2.0 nm (from filter); lamp uncertainty is much smaller

Sources: *The Photoelectric Effect,* MIT Department of Physics, [Wavelengt](#) [spectral lines in mercury pencil lamps](#),

PE Apparatus: Leybold Photocell

Cathode:
potassium; source of photoelectrons **(ϕ ~ 2.3 eV)**



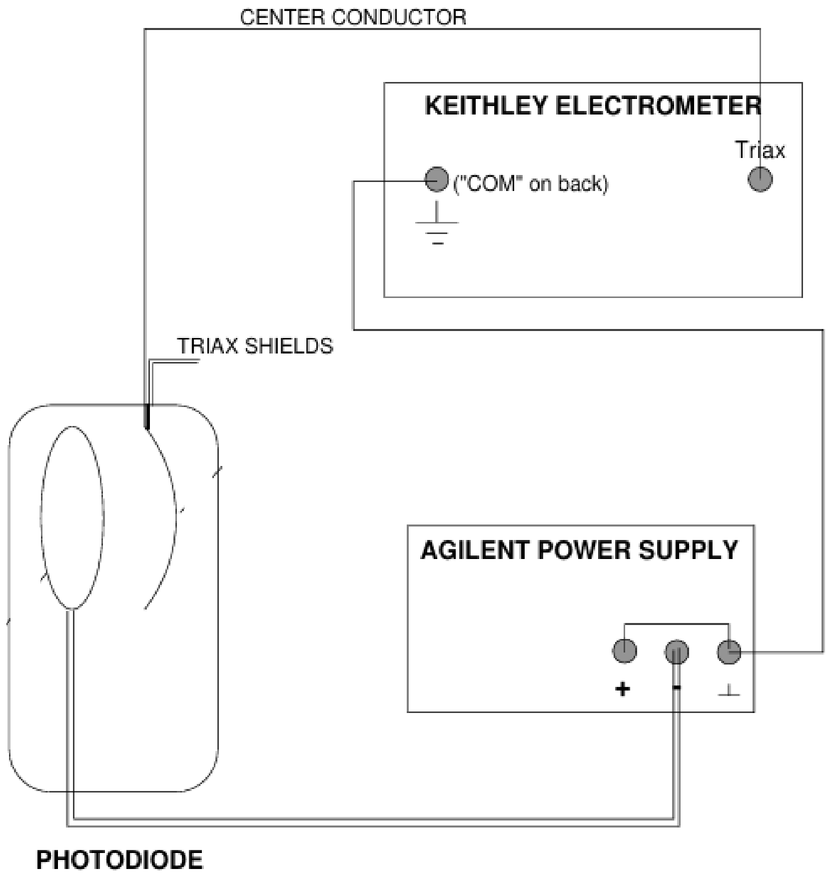
Anode:
Platinum-rhodium alloy; **(ϕ ~ 5.7 eV)**

- **Experimental considerations**
  - Geometry of electron scattering
  - Surface characteristics/variation of work functions
  - Cathode deposition onto anode
  - Reverse current, ambient light

Image credit: https://www.leybold-shop.com/media/phk/images/150dpi/55877.jpg

Sources: *The Photoelectric Effect,* MIT Department of Physics

PE Apparatus: Electrometer and Power Supply



Kiethley Electrometer: connects cathode to ground to measure outgoing photocurrent

- Sensitive to ambient magnetic fields and charges (e.g. cords, hand movements)

Agilent Power Supply: positive (retarding) voltage from cathode to anode, repelling photoelectrons

- Precision: ~0.01 V

Sources: *The Photoelectric Effect,* MIT Department of Physics

Experimental Procedure

1. **Set up the apparatus;** zero-check the ammeter, connect ground, adjust filter orientation

2. **Cover apparatus with black cloth** to filter out ambient light

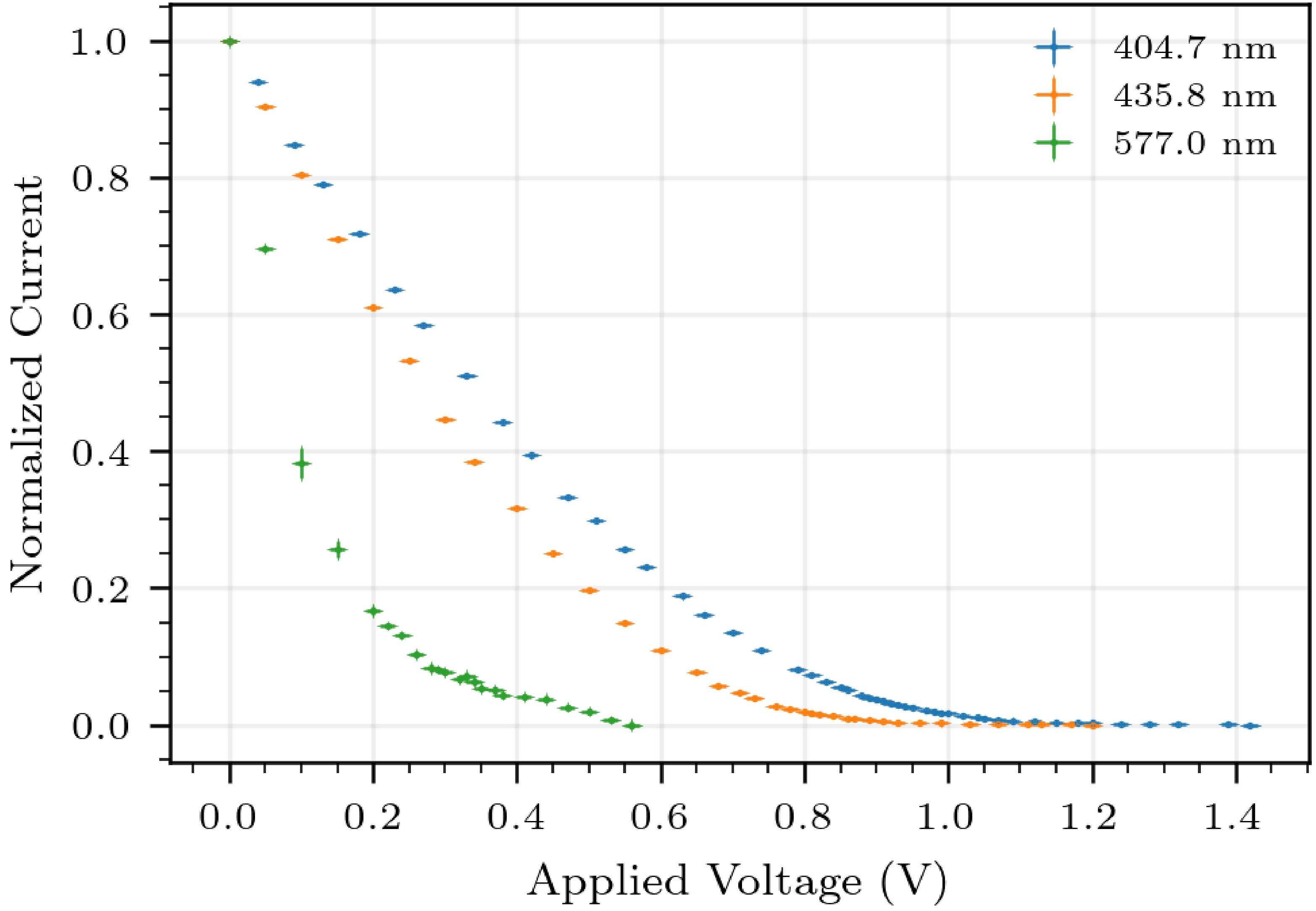
3. For each selected wavelength:

1. Starting from V = 0.00, adjust the power supply voltage in small increments (0.01 to 0.1)
2. Wait ~5 seconds for the current to adjust to new voltage
3. Film ~5 seconds of electrometer current readings
4. Repeat steps 1-3 until current appears to plateau

4. **Take 5 “random” current readings** per voltage for each wavelength

Experimental Results and Analysis

Initial data shows wavelength-dependent IV trends



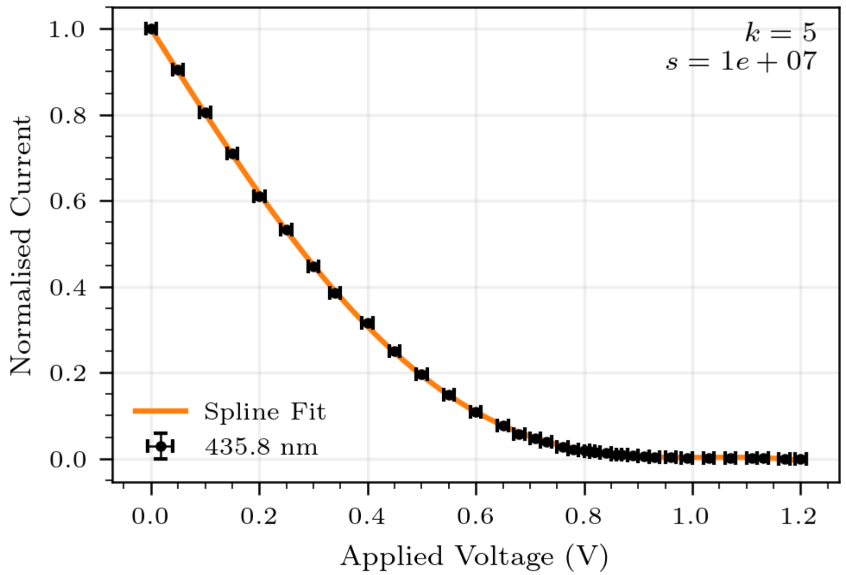
- Currents normalized by magnitude and range:

$$I_n(V) = \frac{I(V) - I_{\min}}{I_{\max} - I_{\min}}$$

- Nonnegligible current appears in measurement time
- Where the current decays depends strongly on wavelength

Spline-fit Bootstrapping method of estimating stopping voltages

- Interpolate smooth function
- Depends on **k** (degree of polynomial, either 4 or 5) and **s** (smoothing parameter)



Stopping Criterion: look for concavity change to indicate stopping potential

$$\frac{d^2 \tilde{I}}{dV^2}(V_s) = 0$$

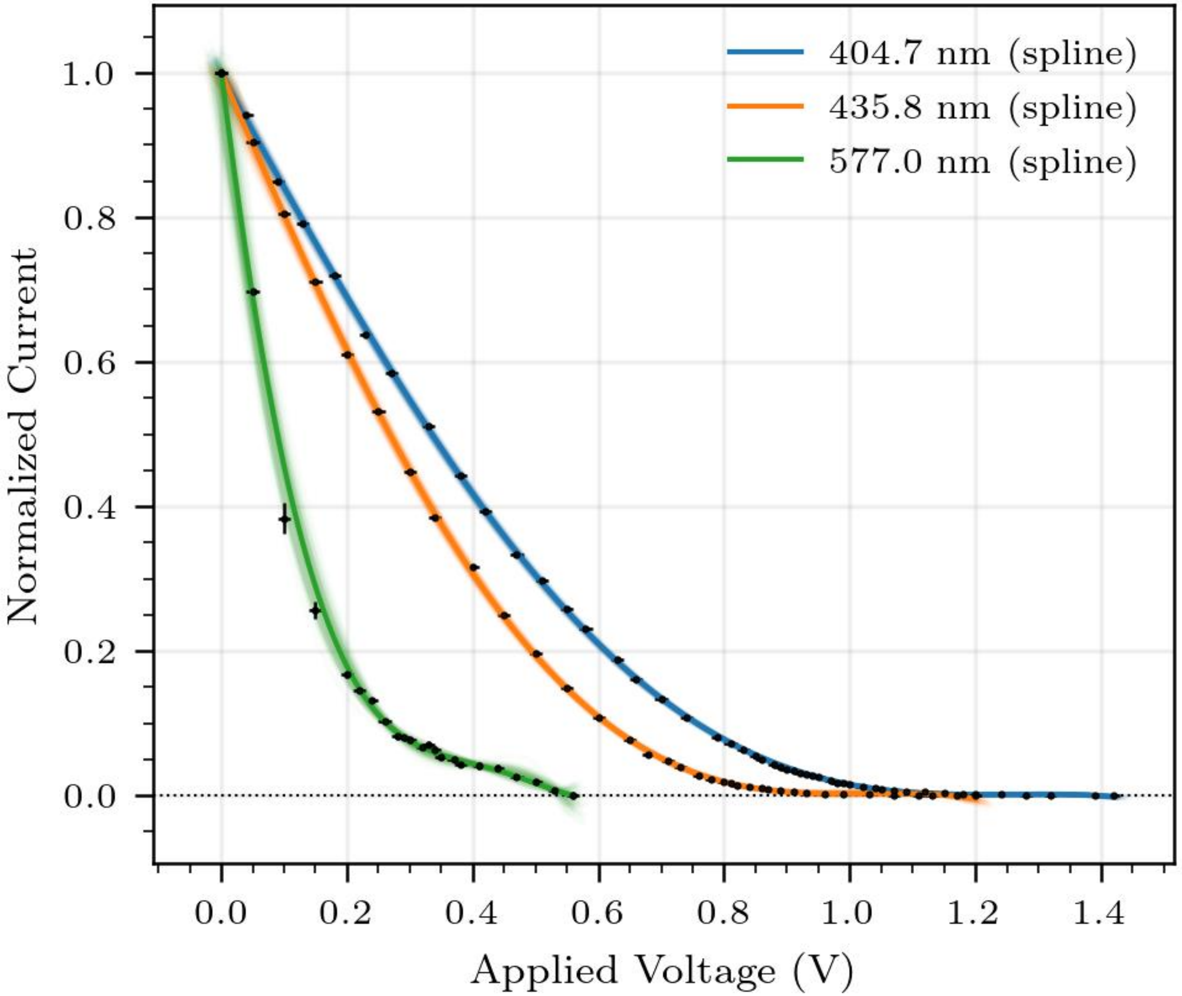
- **Bootstrapping**

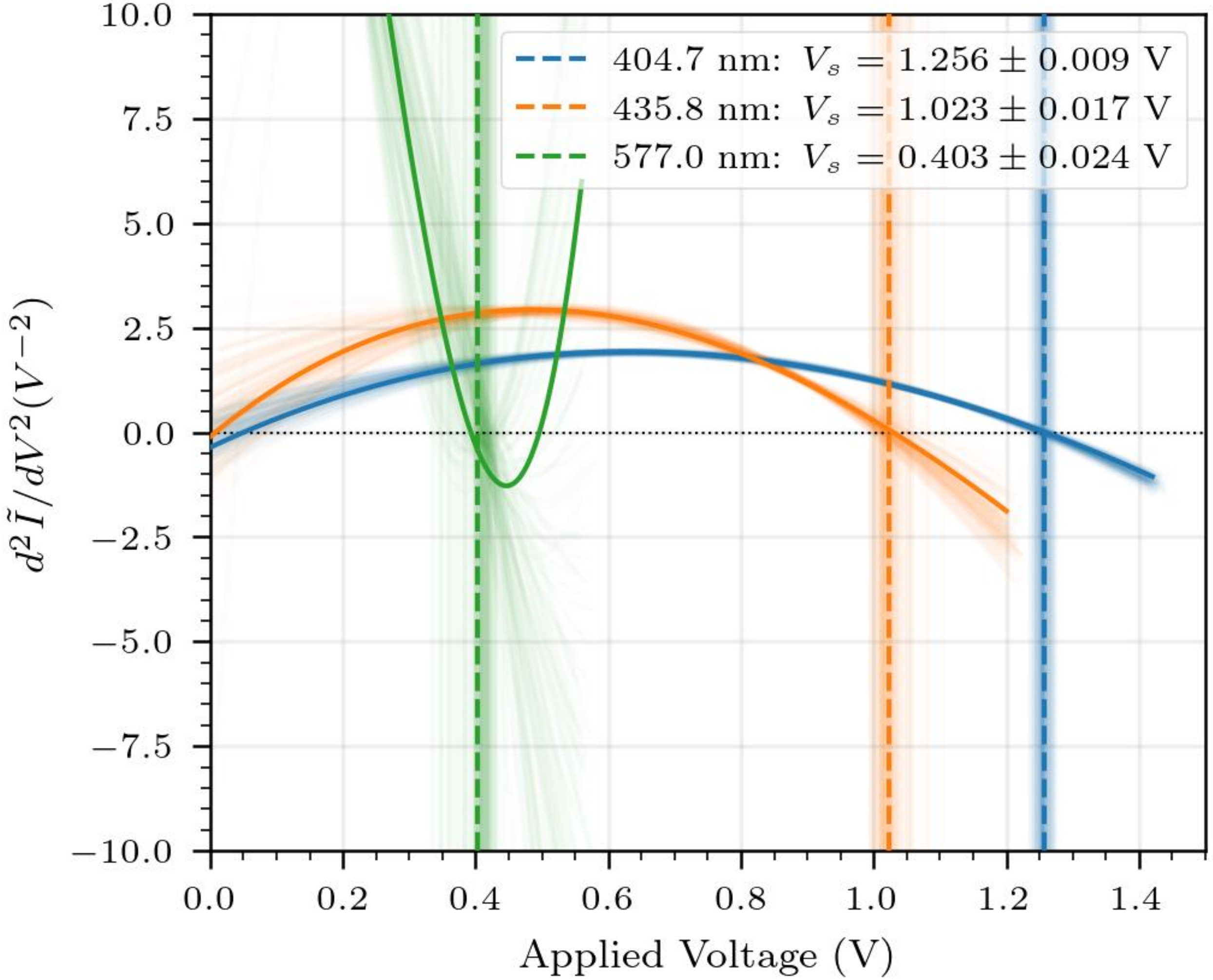
$$V_i^{(b)} = V_i + \varepsilon_i^V, \qquad \varepsilon_i^V \sim \mathcal{N}(0, \sigma_V)$$

$$I_i^{(b)} = I_i + \varepsilon_i^I, \qquad \varepsilon_i^I \sim \mathcal{N}(0, \sigma_I)$$

Sources: https://docs.scipy.org/doc/scipy/reference/generated/scipy.interpolate.UnivariateSpline.html, https://en.wikipedia.org/wiki/Spline_interpolation

Spline-fit Bootstrapping method yields reasonable stopping voltages and statistical uncertainties



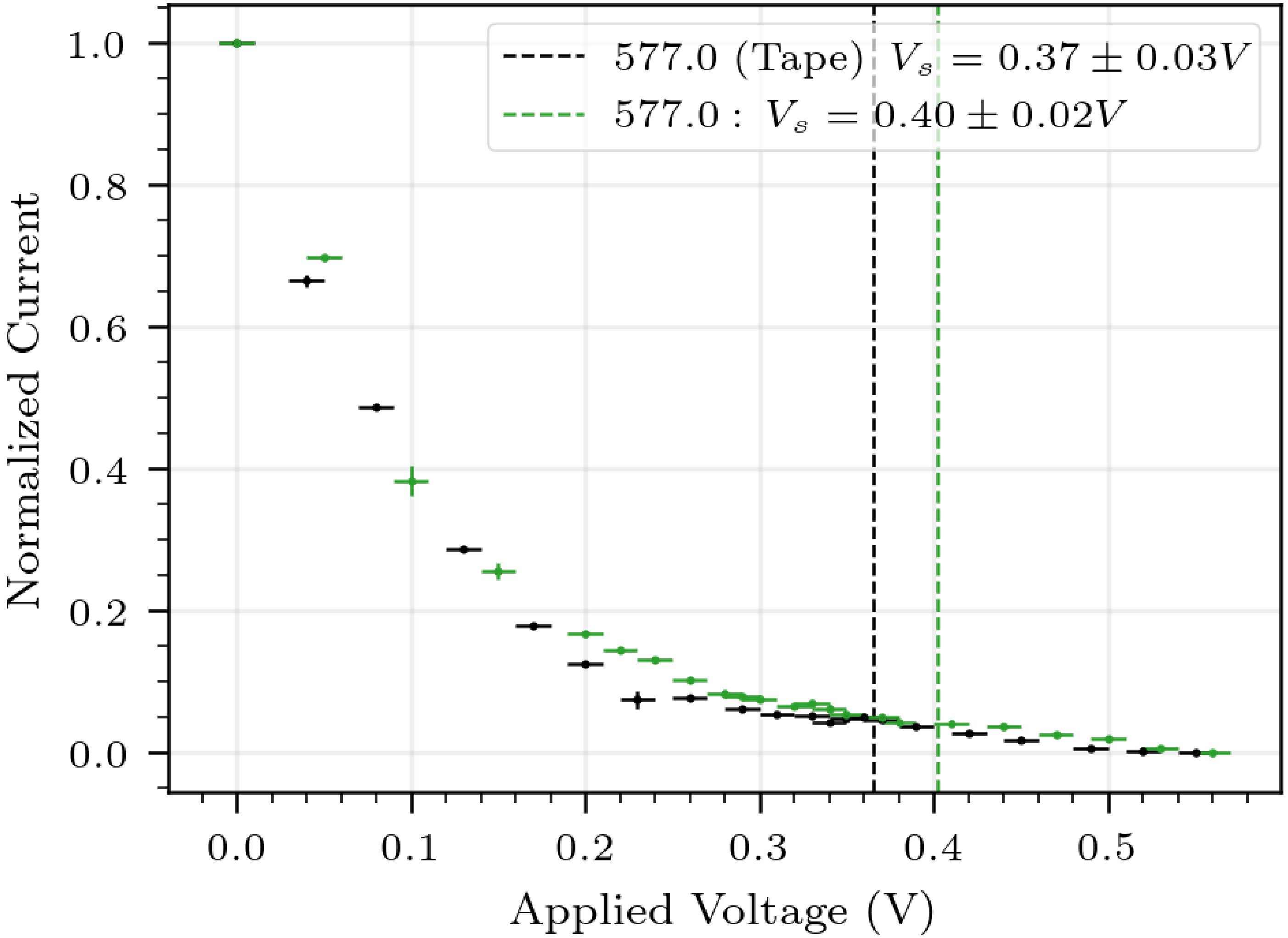


Systematic Error Contributions

- **Thermal Noise** (3.5kbT ~ 0.1 eV), so contribute systematic uncertainty of 0.05 eV in same direction to each stopping voltage
- **Ambient Light, Dark current**
  - We found dark current to be relatively negligible due to normalization, and are taking the second derivative; additionally, black cloth addresses ambient light
- **Model Uncertainty (k=4 or k=5):** Take mean value for h and statistical error; report standard error as systematic error contribution
- **Higher-frequency Light Contamination:** Gap between filter and photocell aperture allows higher-energy light than the set filter get in, broadening the tail

Applying tape decreases the cutoff voltage for 577.0 nm

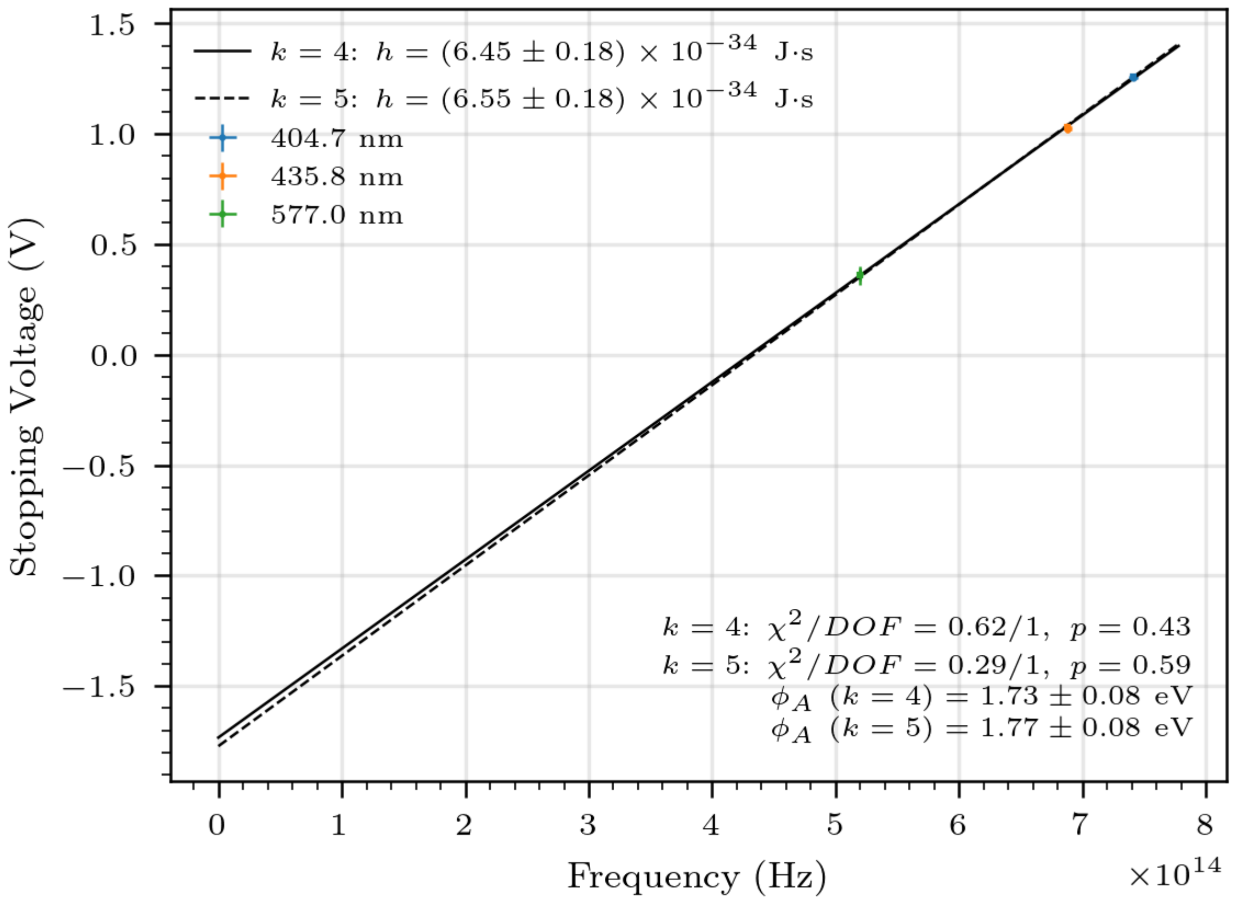
Gap between filter and photocell aperture allows higher-energy light than the filter, broadening the IV tail



Planck's constant near 1σ of accepted value

$$\hat{h} = (6.49 \pm 0.12 \text{ (stat)} \pm 0.05 \text{ (sys)}) \times 10^{-34}\ J \cdot s \qquad h \approx 6.626 \times 10^{-34}\ \mathrm{J \cdot s}$$

$$\hat{\phi}_A = (1.75 \pm 0.06 \text{ (stat)} {}^{+0.02}_{-0.05} \text{ (sys)}) \times 10^{-34}\ \mathrm{eV} \qquad \phi_A \approx 5.7\ \mathrm{eV}$$



- Planck's constant error from expected value: 1.0 σ
- Accuracy would likely improve if tape is added to all wavelengths
- Anode work function error (from 5.7): 49 σ; **Huge systematic shifts not being accounted for**

Other systematic effects this experiment did not account for

Systematic error, especially for the work function, is hugely underestimated

- **Cathode surface variation** in work function
- **Electron scattering geometry** (nonlinear travels paths)
- **Wavelength filter uncertainty** (how much higher-energy photons does it let through?)
- **Cathode deposition** onto anode: mentioned by Melissinos as heavily decreasing anode work function

Conclusion: Photoelectric effect experiment supports Einstein's theory, but inaccurately calculates stopping voltages and work function

- **Evidence for Einstein's theory of light**: Current changes quickly with applied voltage, stopping voltage depends heavily on wavelength
- **Spline-fitting method** consistent across wavelengths (good for finding Planck's constant), but fails at finding accurate energy or voltage values
- **Using tape** to cover tube-filter gap increase h accuracy
- Large systematic errors dominate work function estimate

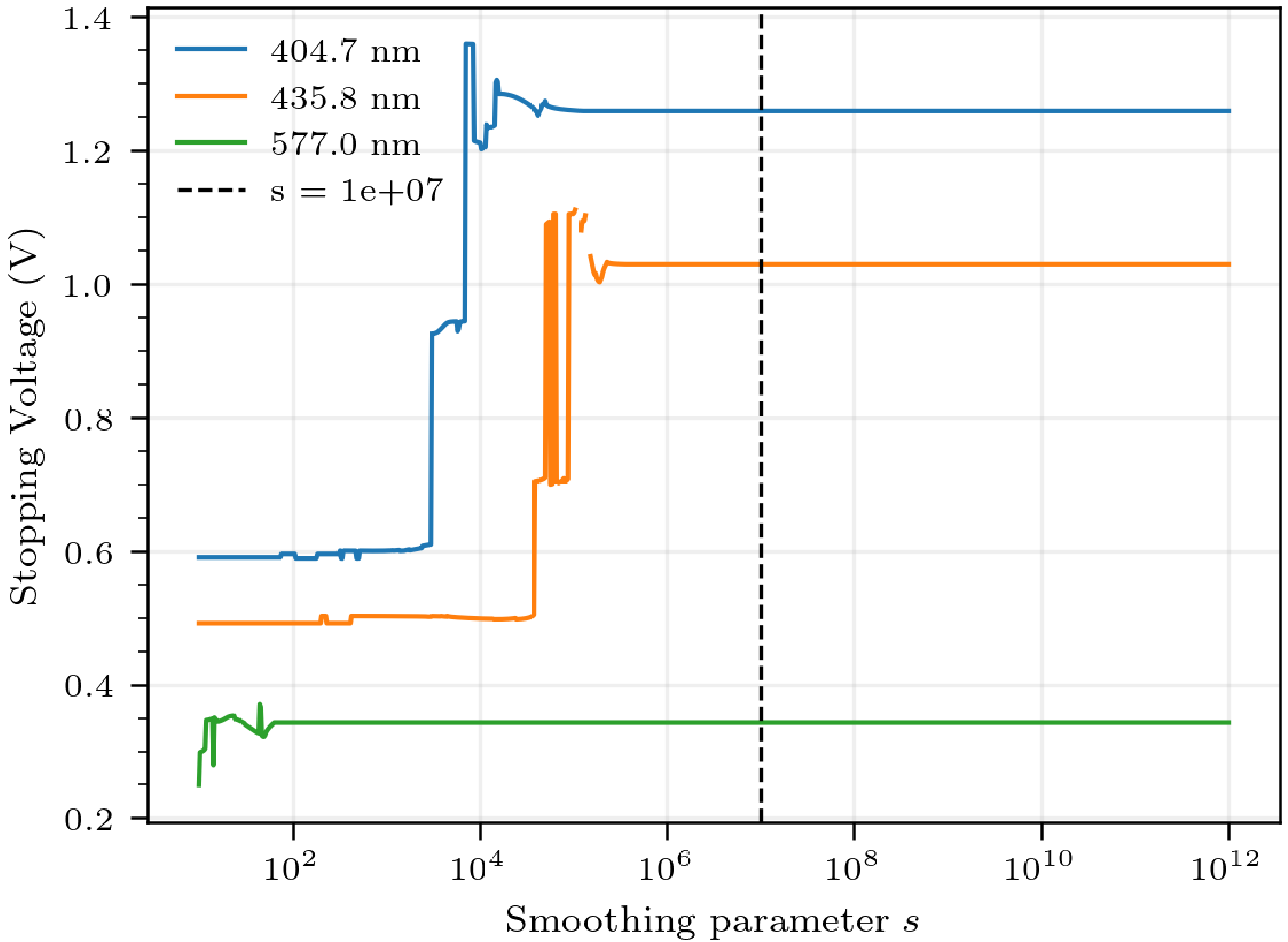
$$E = h\nu$$

$$eV = h\nu - \phi_A$$

Thank you!

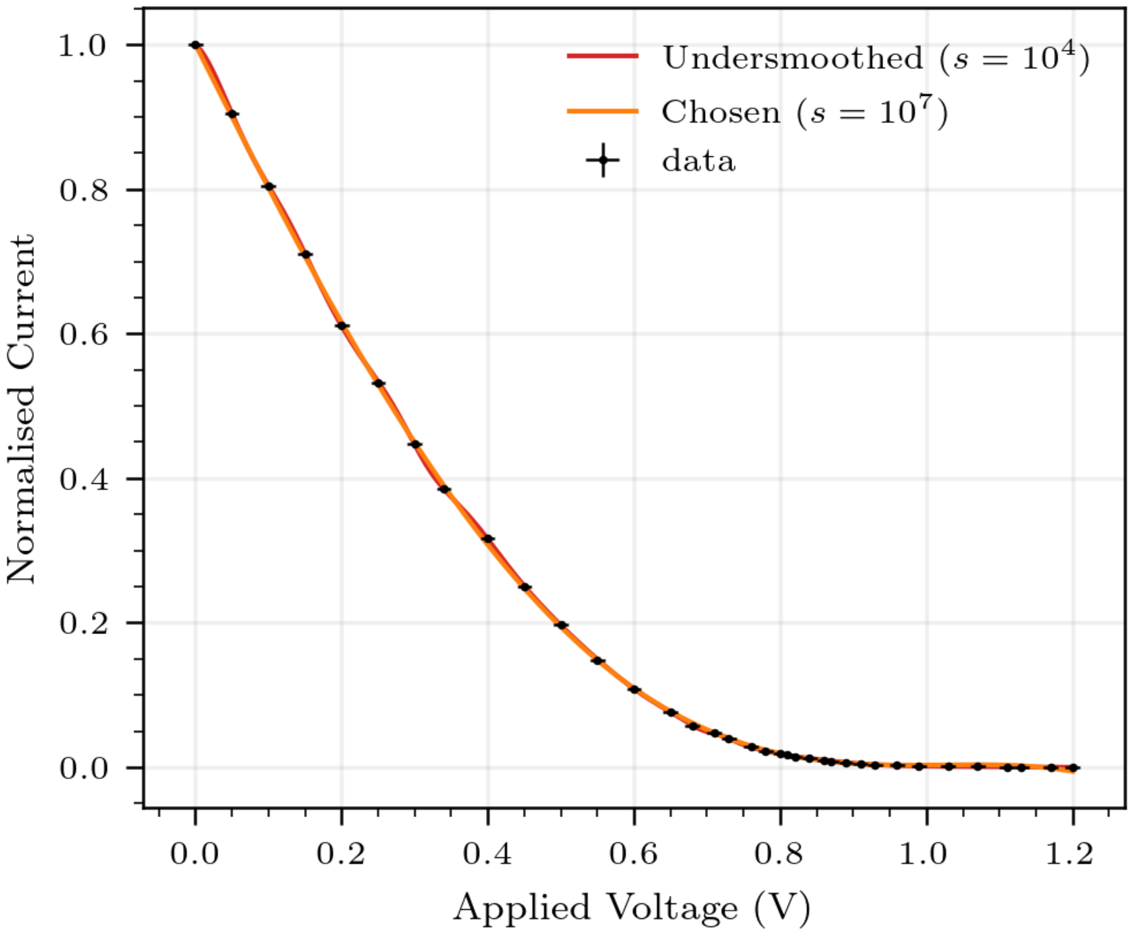
- **Lab Partner:** Vicki Mu
- **Jlab Instructional Staff:** Prof. Janet Conrad, Jixiang Yang, Simon Rothman, Vinh Tran, Atissa Banuazizi
- **JLab Technical Staff:** Dr. Sean Robinson, Dr. Aaron Pilarcik, Cory Romanov
- **Claude Code** and **Github Copilot** were used to generate code for data analysis

Stopping voltages stable for large smoothing parameter

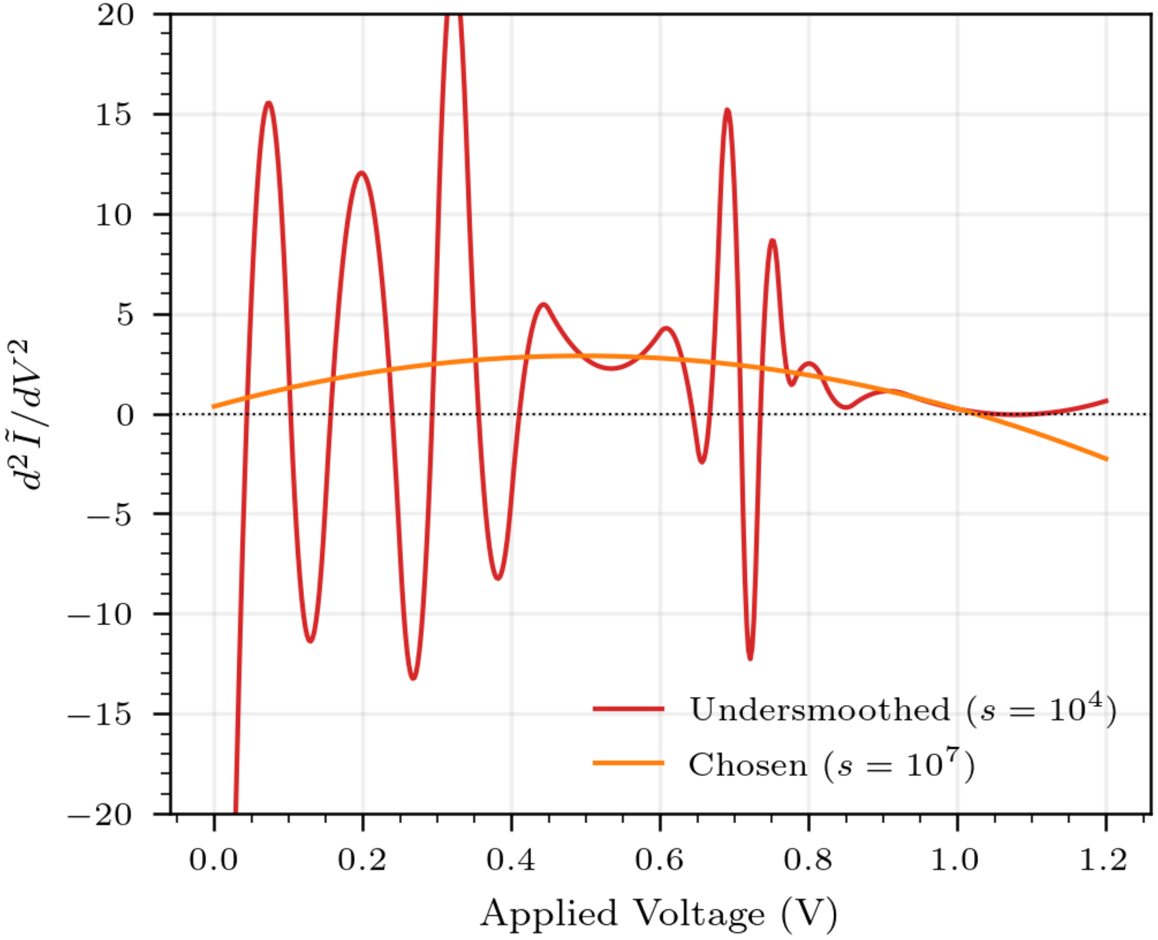


Large smoothing more reasonably captures expected derivative behavior

Spline fit comparison

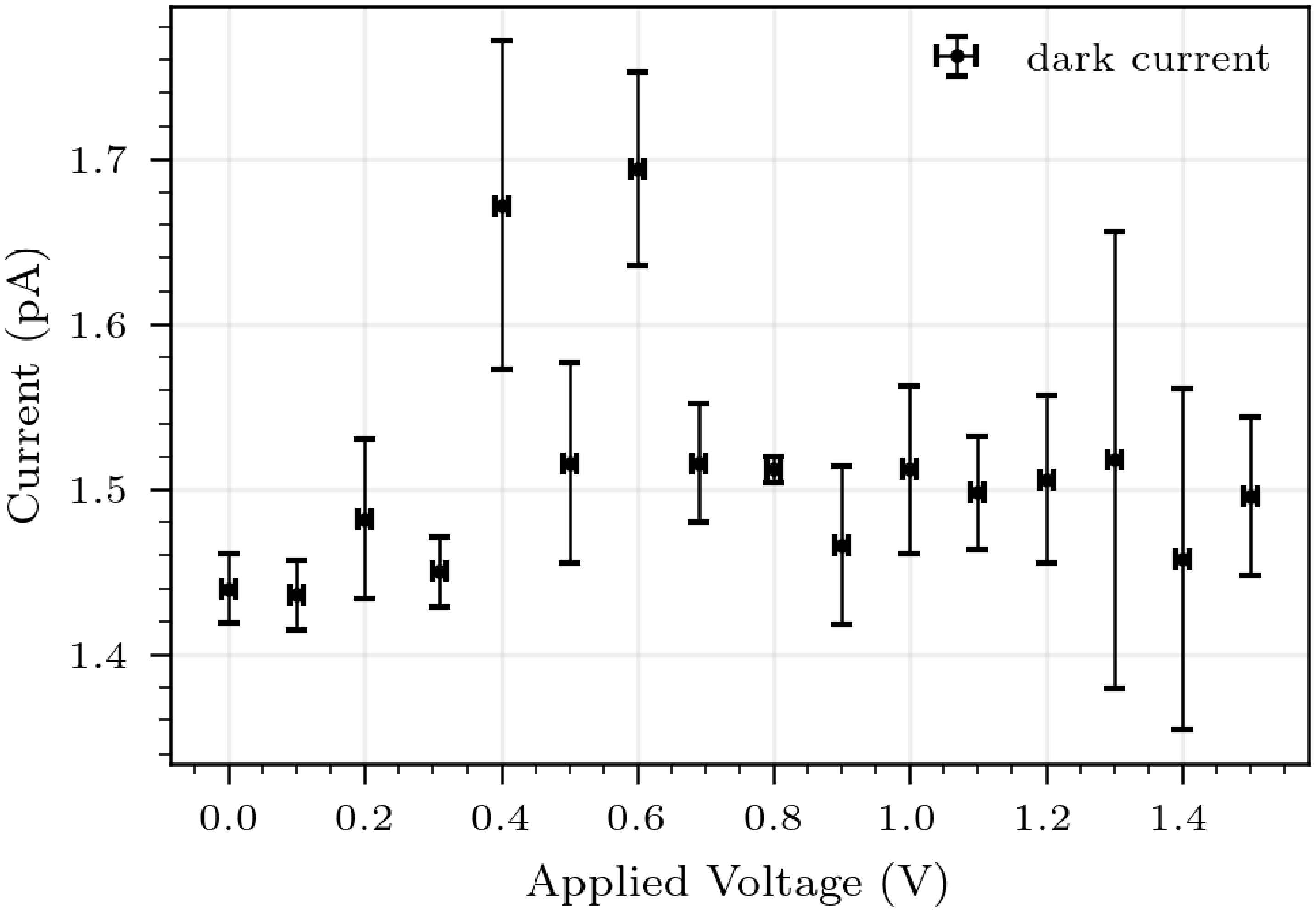


Second Derivative Comparison



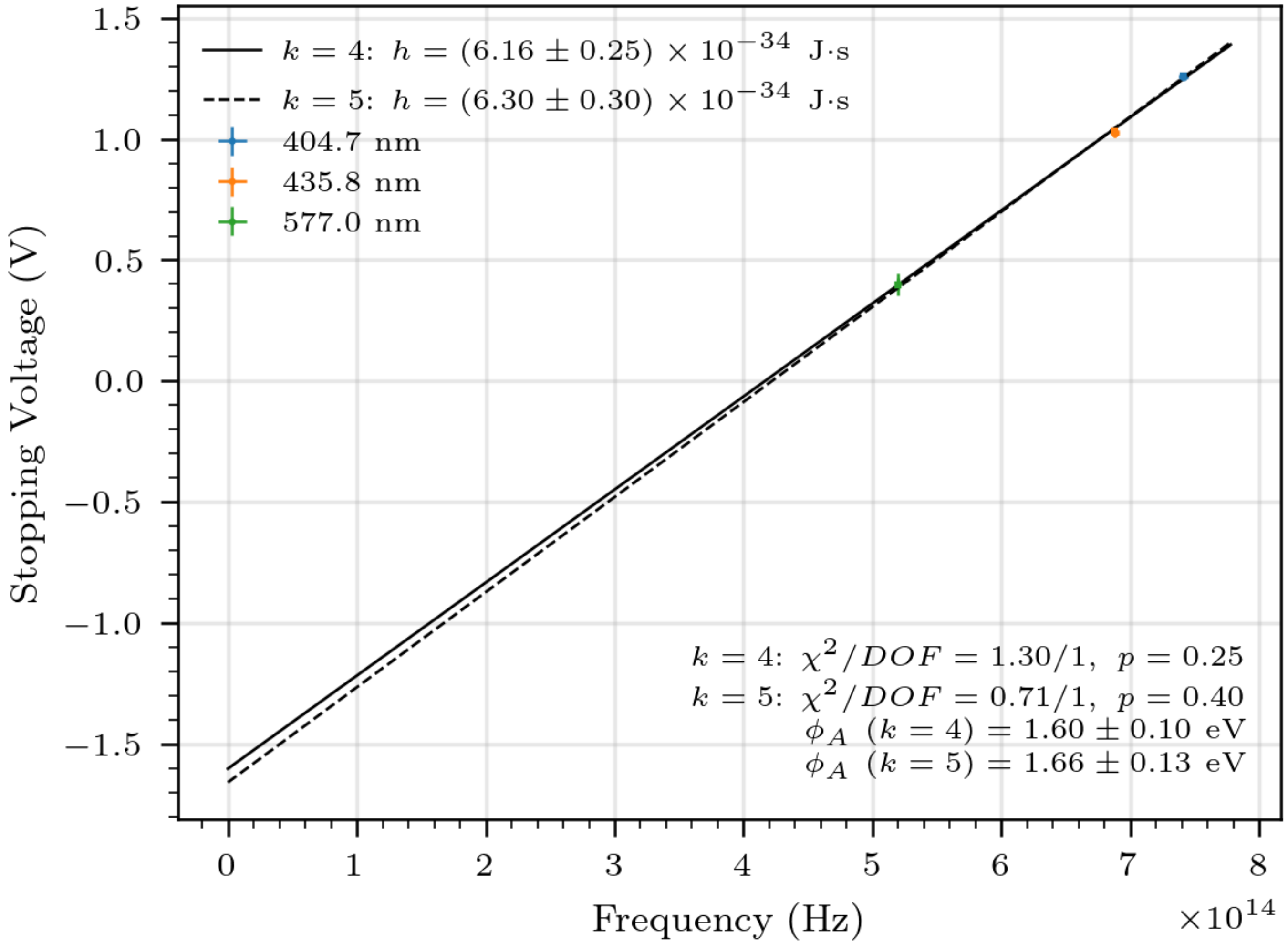
Dark Current Measurement

Current that the sensitive picoammeter picks up that was not due to photoelectron ejection



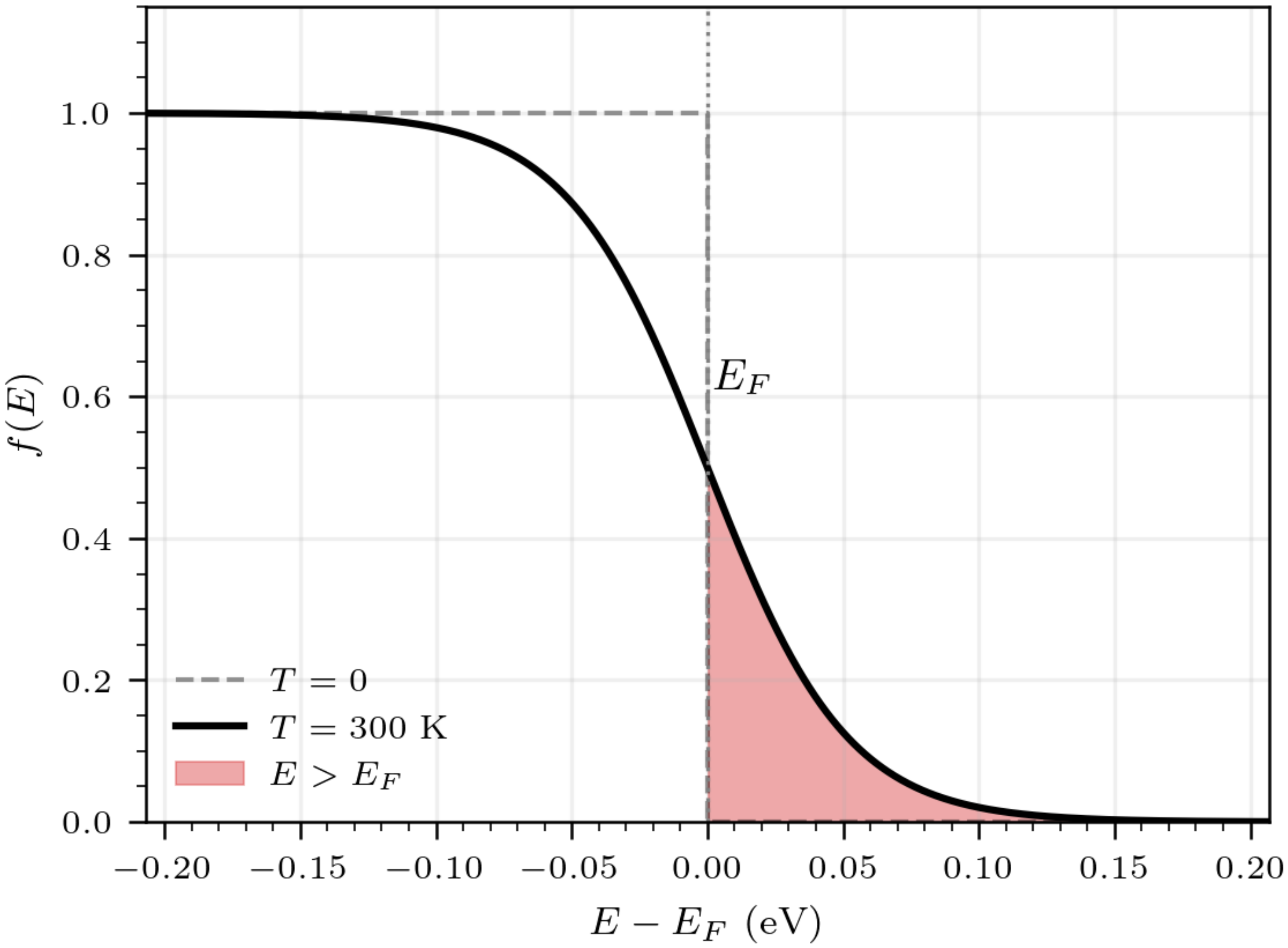
Initial determination of h and phi with statistical uncertainty

$$h = (6.23 \pm 0.19\ (\text{stat})) \pm 0.06\ (\text{sys}) \times 10^{-34}\ J \cdot s$$



- Different h values combined, treating as independent measurements
- Systematic uncertainty calculated as half the difference between model results

Accounting for thermal noise



Uncertainty Flowchart